

Bukti Penyelesaian

SKILL TEST 2

Warehouse Management System — Backend

Work Order · Asset Label · Print PDF · FIFO · Report · Analytics · Export Excel

Tanggal	11 Juni 2026
Project	wms-backend
Stack	Node.js + Express + Prisma v7 + Supabase + JWT
Production URL	https://wms-backend-ef8b.onrender.com/api/v1
Database	Supabase PostgreSQL (aws-1-ap-northeast-1)

1. Ringkasan Requirement Skill Test 2

Berikut adalah seluruh requirement Skill Test 2 beserta status penyelesaiannya:

No	Requirement	Keterangan	Status
1	Menu Operational Inbound Asset (include: Print Label, Scan Label) → MVP	Work Order INBOUND + Generate Labels + Scan Label + Print PDF QR	✓ SELESAI
2	Menu Operational Outbound Asset (include: Scan Label, FIFO) → MVP	Work Order OUTBOUND + Scan Outbound + FIFO Enforcement	✓ SELESAI
3	Analytics: Summary Total Stock All Asset → MVP	GET /reports/analytics dengan totalStock, perCategory, detail	✓ SELESAI
4	Analytics: Summary Total Stock Asset Per Warehouse & Storage Bin	Filter ?warehouseId= dan ?storageBinId=	✓ SELESAI
5	Report: Log Transaction Inbound - Outbound	GET /reports/inbound dan /reports/outbound	✓ SELESAI
6	Report: Download Excel Log Transaction Inbound - Outbound - Stock	GET /reports/export/excel → download .xlsx (3 sheet: Inbound, Outbound, Stock)	✓ SELESAI

2. Daftar Endpoint yang Diimplementasikan

2.1 Work Order

Method	Endpoint	Deskripsi	Status
GET	/work-orders	List semua Work Order	✓ PASS
GET	/work-orders/:id	Detail Work Order + labels	✓ PASS
POST	/work-orders	Buat Work Order baru	✓ PASS

Method	Endpoint	Deskripsi	Status
PUT	/work-orders/:id/status	Update status WO	✓ PASS
POST	/work-orders/:id/generate-labels	Generate label sesuai qty	✓ PASS
GET	/work-orders/:id/fifo-labels	Ambil label urutan FIFO	✓ PASS

2.2 Asset Label

Method	Endpoint	Deskripsi	Status
GET	/asset-labels	List semua label	✓ PASS
GET	/asset-labels/:id	Detail label + scans	✓ PASS
POST	/asset-labels/scan	Scan label inbound	✓ PASS
POST	/asset-labels/outbound-scan	Scan label outbound (FIFO)	✓ PASS
GET	/asset-labels/print/:workOrderId	Download PDF label + QR Code	✓ PASS

2.3 Report & Analytics

Method	Endpoint	Deskripsi	Status
GET	/reports/inbound	Report transaksi inbound	✓ PASS
GET	/reports/outbound	Report transaksi outbound	✓ PASS
GET	/reports/stock	Report stock semua asset	✓ PASS
GET	/reports/analytics	Analytics summary + filter warehouse	✓ PASS
GET	/reports/export/excel	Download Excel (.xlsx) — 3 sheet: Inbound Log, Outbound Log, Stock Summary	✓ PASS

3. Bukti Testing di Postman (Production)

Semua testing dilakukan pada server production Render.com dengan database Supabase PostgreSQL.

Step 0 — Login & Mendapatkan Token

Endpoint	POST /auth/login
Akun	admin@wms.com / Admin123! (Role: ADMIN)
Result	✓ Token JWT berhasil didapatkan — success: true, message: Login successful

Step 1 — Work Order Create (INBOUND)

Endpoint: POST /work-orders

type	INBOUND
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warehouseId	cmp9h1bg5000104v5gefaf9ik (Gudang Jogja - WH_01)
storageBinId	cmp9j1x830002qov5ryk0tbxy (WH_01_003)
assetId	cmpdsv8zf00091shw9gq4nmda (AST_04 - Barang A)
quantity	3
remarks	Test inbound skill test 2
Response	✅ woNumber: WOI_002 status: TODO id: cmq8vcq2z00002add0urili3b

Step 2 — Generate Labels

Endpoint: POST /work-orders/cmq8vcq2z00002add0urili3b/generate-labels

Label	Label Code	Inbound At
Label 1	AST_04_000001	2026-06-11T02:19:41.250Z
Label 2	AST_04_000002	2026-06-11T02:19:41.559Z
Label 3	AST_04_000003	2026-06-11T02:19:41.865Z
Verifikasi		

Step 3 — Scan Label Inbound (3x)

Endpoint: POST /asset-labels/scan

Scan	Label Code	Status WO	Total Scan
Scan ke-1	AST_04_000001	ON_PROGRESS	1 / 3
Scan ke-2	AST_04_000002	ON_PROGRESS	2 / 3
Scan ke-3	AST_04_000003	DONE	3 / 3
Verifikasi			

Step 4 — Print Label PDF

Endpoint: GET /asset-labels/print/:workOrderId

Format PDF	A4, 2 kolom × 5 baris = 10 label per halaman
Isi Label	Asset Number, Asset Name, Harga, QR Code, Label Code, Supplier
QR Content	Label Code (AST_04_000001, AST_04_000002, AST_04_000003)
Hasil	✅ PDF berhasil didownload 3 label dengan QR Code muncul Format label sesuai requirement

Step 5 — Work Order Create (OUTBOUND)

Endpoint: POST /work-orders

Response	✅ woNumber: WOO_001 type: OUTBOUND status: TODO quantity: 2
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Step 6 — FIFO Labels Suggestion

Endpoint: GET /work-orders/:id/fifo-labels

Urutan	Label Code	Inbound At	Status
1	AST_04_000001	2026-06-11T02:19:41.250Z (terlama)	isOutbound: false
2	AST_04_000002	2026-06-11T02:19:41.559Z	isOutbound: false
Verifikasi			

Step 7 — Scan Label Outbound (FIFO Enforcement)

Endpoint: POST /asset-labels/outbound-scan

Skenario	Label Di-scan	Hasil	Keterangan
Scanurut FIFO ke-1	AST_04_000001	✓ Sukses	isOutbound: true
Scanurut FIFO ke-2	AST_04_000002	✓ Sukses	isOutbound: true
Scan label sudah outbound	AST_04_000004	✓ Error 400	Label already outbound
Verifikasi FIFO			

Step 8 & 9 — Report Inbound & Outbound

Report	Endpoint	Hasil
Inbound	GET /reports/inbound	✓ 7 record — woNumber, labelCode, scannedAt, scannedBy, updatedStock
Outbound	GET /reports/outbound	✓ 5 record — woNumber, labelCode, outboundAt

Step 10 — Report Stock

Endpoint: GET /reports/stock

Asset No	Asset Name	Storage Bin	Stock	Harga
AST_01	Nike Journey Run...	-	0	Rp 100.000
AST_02	Nike Journey Run...	WH_01_004	9	Rp 100.000
AST_03	Nike Journey Run...	WH_01_001	10	Rp 100.000
AST_04	Barang A	WH_01_002	0	Rp 10.000
AST_09	Nike Journey Run	-	1	Rp 1.200.000
... dst	Total 9 Asset	-	20	-

Step 11 & 12 — Analytics (Semua Asset + Filter Warehouse)

Query	Response	Hasil
GET /reports/analytics (semua)	totalAsset: 8, totalStock: 20, perCategory: SMALL_ASSET	✅ Analytics lengkap dengan detail per asset
GET /reports/analytics?warehouseId=... (Gudang Jogja)	totalAsset: 3, totalStock: 19, hanya asset di WH_01	✅ Filter warehouse bekerja dengan benar

Step 13 — Export Excel (NEW)

Endpoint: GET /reports/export/excel

Library	exceljs (npm install exceljs)
Format Output	.xlsx — Microsoft Excel / Google Sheets compatible
Sheet 1	Inbound Log — WO Number, WO Category, Warehouse, Storage Bin, Asset Name, Supplier, Remarks, Label Code, Scanned At, Scanned By, Updated Stock
Sheet 2	Outbound Log — WO Number, WO Category, Warehouse, Storage Bin, Asset Name, Supplier, Label Code, Inbound At, Outbound At
Sheet 3	Stock Summary — Asset Number, Asset Name, Category, Storage Bin, Warehouse, Supplier, Stock, Price
Header Style	Bold + colored background per sheet (Navy/Green/Blue)
Hasil	✅ File .xlsx berhasil didownload dari Postman 3 sheet muncul sesuai requirement Data real dari database production

4. Bug yang Ditemukan dan Diperbaiki

No	Lokasi	Sebelum (Bug)	Sesudah (Fix)
1	assetLabelController.js — scanLabel	scannedBy: req.user.id	scannedById: req.user.id (nama field di schema)
2	assetLabelController.js — scanLabel	status = 'IN_PROGRESS'	status = 'ON_PROGRESS' (sesuai enum schema)
3	workOrder.js (route)	Tidak ada POST /	Ditambah POST / + validate middleware
4	reportRoutes.js	Route tanpa authenticate	Semua route dilindungi middleware authenticate
5	assetLabelController.js — scanLabel	inboundAt diset saat generate label (timestamp sama)	inboundAt diset saat scan (akurat untuk urutan FIFO)

5. Rekap Hasil Testing

Step	Fitur	Yang Diverifikasi	Status
0	Login & Token	JWT token berhasil didapatkan	✅ PASS
1	Work Order INBOUND	woNumber WOI_002, status TODO	✅ PASS
2	Generate Labels	3 label format AST_04_XXXXXX	✅ PASS

Step	Fitur	Yang Diverifikasi	Status
3	Scan Inbound	Status TODO → ON_PROGRESS → DONE, stock +3	✓ PASS
4	Print PDF + QR Code	PDF A4 dengan QR Code per label	✓ PASS
5	Work Order OUTBOUND	woNumber WOO_001, status TODO	✓ PASS
6	FIFO Labels	Urut dari inboundAt terlama	✓ PASS
7	Scan Outbound FIFO	FIFO enforcement + duplicate ditolak	✓ PASS
8	Report Inbound	Data scan inbound lengkap	✓ PASS
9	Report Outbound	Data label outbound lengkap	✓ PASS
10	Report Stock	Semua asset dengan stock terkini	✓ PASS
11	Analytics Semua	totalStock, perCategory, detail	✓ PASS
12	Analytics Filter	Filter per warehouseId bekerja	✓ PASS
13	Export Excel	3 sheet: Inbound Log, Outbound Log, Stock Summary — file .xlsx berhasil didownload	✓ PASS

Semua requirement MVP Skill Test 2 telah berhasil diimplementasikan dan diuji pada environment production.

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